

VHC COMMUNICATION MESSAGE SPECIFICATION

Version	Document			Model
1.0.0	VHC COMMUNICATION BASIC SPECIFICATION			
Date	Drawn	Checked	Approved	Page

REVISION HISTORY

VER	DATE	DESCRIPTION	Note
2.2	2018-03-05	Cassette Type added to S2F49 and S6F11(RPTID = 4). Port Type added to Current Port States of S1F4. Scan Command Scenario added.	
2.3	2018-03-07	Install / Remove Command(S2F41) added VehicleCarrierInstall/Remove RPTID=16 CEID 153,154 added.	
2.5	2018-03-10	SCAN Command changed. CarrierLoc delete from RPTID 4(CarrierInstalled/Removed) CarrierLoc added to RPTID 11.	
2.6	2018-03-28	Cassette Position added - S1F4 - RPTID 16	
3.0	2018-09-10	WaitIn / WaitOut Added.	
3.2	2018-12-19	대만 마이크론 L10 S1F13 / S2F15 수정 및 CarrierType 추가	
3.3	2019-01-02	Abort Command 수정	
3.4	2019-02-23	VehicleCarrierInstall/Remove rollback	

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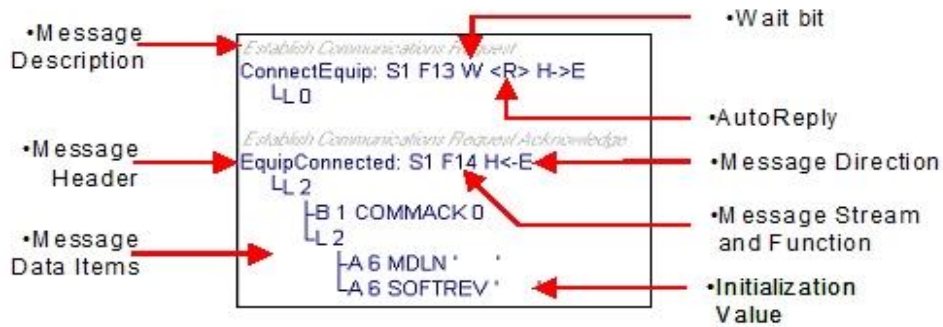
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1. SECS-II MESSAGE AND DESCRIPTIONS

1.1. SECS-II Message and Descriptions

SECS-II	Description
Stream 1	Equipment Status
Stream 2	Equipment Control and Diagnostics - Host sends equipment to Constants, defines trace reporting, define reports and links events to reports, and sends commands to equipment
Stream 3	Material Status
Stream 4	Material Control - Request to send and receive materials
Stream 5	Exception Handling and Alarms Communications
Stream 6	Data Collection - Equipment sends Trace Data and Event Reports to Host
Stream 7	Process Program Management and Communication
Stream 8	Boot Program Request
Stream 9	System Errors - Error messages sent from the equipment to the host
Stream 10	Terminal Requests
Stream 11	Host File Services – Deleted from Standard
Stream 12	Wafer Mapping Data Sent
Stream 13	Data Set Transfer
Stream 14	Object Services
Stream 15	Recipe management, DRNS
Stream 16	Process Job Communications
Stream 17	Equipment Control and Diagnostics; Trace Reporting
Stream 18	Subsystem Control and Data

1.2. SECS-II Message Format



1.3. Supported Stream and Function List

Message	Direction	Description	Note
S1		EQ Status	
S*F0	Host ↔ EQ	Abort Transaction	
S1F1	Host ↔ EQ	Are You There Request	
S1F2	Host ↔ EQ	On-line Data	
S1F3	Host → EQ	Selected Equipment Status Request	
S1F4	Host ← EQ	Selected Equipment Status Data	
S1F13	Host ↔ EQ	Establish Communications Request	
S1F14	Host ↔ EQ	Establish Communications Request Acknowledge	
S1F15	Host → EQ	Request OFF-LINE	
S1F16	Host ← EQ	OFF-LINE Acknowledge	
S1F17	Host → EQ	Request ON-LINE	
S1F18	Host ← EQ	ON-LINE Acknowledge	
S2		EQ Control	
S2F13	Host → EQ	Equipment Constant Request (FDC Data List Request)	
S2F14	Host ← EQ	Equipment Constant Data (FDC Data List)	
S2F15	Host → EQ	New Equipment Constant Send	
S2F16	Host ← EQ	New Equipment Constant Acknowledge	
S2F17	Host ← EQ	Date and Time Request	

S2F18	Host → EQ	Date and Time Data	
S2F29	Host → EQ	Equipment Constant Name List Request	NO USED
S2F30	Host ← EQ	Equipment Constant Name List	NO USED
S2F31	Host → EQ	Date and Time Set Data	
S2F32	Host ← EQ	Date and Time Set Acknowledge	
S2F33	Host → EQ	Define Report	NO USED
S2F34	Host ← EQ	Define Report Acknowledge	NO USED
S2F35	Host → EQ	Link Event Report	NO USED
S2F36	Host ← EQ	Link Event Report Acknowledge	NO USED
S2F37	Host → EQ	Enable/Disable Event Report	
S2F38	Host ← EQ	Enable/Disable Event Report Acknowledge	
S2F41	Host → EQ	Host Command Send	
S2F42	Host ← EQ	Host Command Acknowledge	
S2F49	Host → EQ	Enhanced Remote Command	
S2F50	Host ← EQ	Enhanced Remote Command Acknowledge	
S5		Alarm	
S5F1	Host ← EQ	Alarm Report Send	
S5F2	Host → EQ	Alarm Report Acknowledge	
S5F3	Host → EQ	Enable/Disable Alarm Send	
S5F4	Host ← EQ	Enable/Disable Alarm Acknowledge	
S5F5	Host → EQ	Alarm List Request	
S5F6	Host ← EQ	Alarm List Data	
S6		Data Collection	
S6F11	Host ← EQ	Event Report Send	
S6F12	Host → EQ	Event Report Acknowledge	
S6F15	Host → EQ	Event Report Request	NO USED
S6F16	Host ← EQ	Event Report Data	NO USED
S9		Abnormal	

S9F1	Host $\leftarrow$ EQ	Unrecognized Device ID	
S9F3	Host $\leftarrow$ EQ	Unrecognized Stream Type	
S9F5	Host $\leftarrow$ EQ	Unrecognized Function Type	
S9F7	Host $\leftarrow$ EQ	Illegal Data	
S9F9	Host $\leftarrow$ EQ	Transaction Timer Timeout	

2. SECS-II / GEM MESSAGE DETAIL

2.1. S*F0 (Abort Transaction)

Stream	Function	Description	Wait Bit	Transfer
*	0	Used in lieu of an expected reply to abort a transaction. Function 0 is defined in every stream and has the same meaning in every stream.	0	H ↔ E

1) Definition

None

2) Message Structure

HEADER ONLY.

2.2. S1F1 (Are you there Request)

Stream	Function	Description	Wait Bit	Transfer
1	1	Establishes if the equipment is on-line. A function 0 response to this message means the communication is inoperative. In the equipment, a function 0 is equivalent to a timeout on the receive timer after issuing S1,F1 to the host.	1	H ↔ E

1) Definition

None

2) Message Structure

HEADER ONLY.

MSG:S1F1[Are you there Request]

2.3. S1F2 (On Line Data)

Stream	Function	Description	Wait Bit	Transfer
1	2	Data signifying that the equipment is alive.	0	H ↔ E

1) Definition

None

2) Message Structure

[Case 1 : Equipment → Host]

1. <L [2]

1. <B[1] COMMACK>

2. <L [4]

1. <A[6] MDLN>

* Model Name of the equipment

2. <A[6] SOFTREV>

* Software Revision Number

3. <A[30] EQPNAME>

* EQPNAME

4. <U2 MCMD>

* Online Mode (OFFLINE, ONLINE)

>

>

[Case2 : Host → Equipment]

1. <L [1]

1. <A[6] MDLN> * Model Name of the MCS

>

[Case 1 : Equipment → Host]

MSG:S1F2[Online Data]

<L [2]

<B[1] COMMACK '0'>

<L [4]

<A[6] MDLN 'RCP'>

<A[6] SOFTREV '30101'>

<A[30] EQPNAME 'A2LOHS01'>

<U2 MCMD '1'>

>

>

[Case2 : Host → Equipment]

MSG:S1F13[Establish Communications Request]

<L [1]

<A[3] MDLN 'MCS'>

>

2.4. S1F3 (Selected Equipment Status Request)

Stream	Function	Description	Wait Bit	Transfer
1	3	A request to the equipment to report selected values of its status.	1	H → E

1) Definition

None

2) Message Structure

1. <L [n]

* n = Number of SV

1. <U2 SVID>

* Status Variable ID

:

n. <U2 SVID>

>

* n = 0 means report all SVIDs.

MSG:S1F3[Selected Equipment Status Request]

<L [10]

<U2 [1] 4 >

* Alarm Set

<U2 [1] 53 >

* ActiveVehicle

<U2 [1] 118 >

* CurrentPortState

<U2 [1] 91 >

* EnhancedCarriers

<U2 [1] 6 > * ControlState
 <U2 [1] 73 > * TSCState
 <U2 [1] 76 > * EnhancedTransfers
 <U2 [1] 254 > * UnitAlarmStatList
 <U2 [1] 371 > * CstInfoList
 <U2 [1] 7 > * **EventsEnabled**
 >

2.5. S1F4 (Selected Equipment Status Data)

Stream	Function	Description	Wait Bit	Transfer
1	4	The equipment reports the value of each SVID requested in the order requested. The host remembers the name of values requested.	0	H ← E

1) Definition

SV : Status Variable : Refer to the VID Table

2) Message Structure

1. <L [n] * n = Number of SV
 1. <SVID> * Refer to the VID Table
 :
 n. <SVID> * Refer to the VID Table
 >

* n = 0 is not means that SVID does not exist, but means report all SVIDs

MSG:S1F4[Selected Equipment Status Data]

<L [9]

<L [1] AlarmSets

<U4 1 ALARM(1) '15794431'>

>

<L [n] ActiveVehicles

<L [3] VehicleInfo(1)

<A 13 VEHICLEID(1) 'A2LOHS01_0101'>

<U2 1 VEHICLESTATE(1) '2'>

<A 4 VEHICLELOCATION(1) '1023'>

>

...

<L [3] VehicleInfo(n)

<A 13 VEHICLEID(n) 'A2LOHS01_0109'>

<U2 1 VEHICLESTATE(n) '2'>

<A 4 VEHICLELOCATION(n) '1089'>

>

>

<L [n] CurrentPortStates

<L [2] PortInfo(1)

<A 13 PORTID(1) 'A1STK01D_TI01'>

```

    <U2 1 PORTTRANSFERSTATE(1) '2'>
    <U2 1 PORTTYPE(1) '1'>
  >
  ...
  <L [2] PortInfo(n)
    <A 13 PORTID(n) 'A1STK09D_TO01'>
    <U2 1 PORTTRANSFERSTATE(n) '2'>
  >
  >
  <L [n] EnhancedCarriers
    <L [7] EnhancedCarrierInfo(1)
      <A 8 CARRIERID(1) '1FS6263A'>
      <A 13 VEHICLEID(1) 'A2LOHS01_0101'>
      <A 0 CARRIERLOC(1) ">
      <A 16 INSTALLTIME(1) '2010042119382995'>
      <U2 1 CSTSIZE(1) '2'>
      <A4 CARRIERTYPE(n) 'TYPE'>
      <U2 1 CstPosition(1) '1'>
    >
    ...
    <L [7] EnhancedCarrierInfo(n)
      <A 8 CARRIERID(n) '1FS6265C'>
      <A 6 VEHICLEID(n) 'A2LOHS01_0106'>
      <A 5 CARRIERLOC(n) 'A1STK06D_TI01'>
      <A 16 INSTALLTIME(n) '2010042119382995'>
      <U2 1 CSTSIZE(1) '2'>
      <A4 CARRIERTYPE(n) 'TYPE'>
      <U2 1 CstPosition(1) '1'>
    >
  >
  <U2 1 CONTROLSTATE '5'>
  <U2 1 TSCSTATE '3'>
  <L [n] EnhancedTransfers>
    <L [3]
      <L [3] CommandInfo(1)
        <A 33 COMMANDID(1) '1FS5012B_20100626102925830_MESSYS'>
        <U2 1 PRIORITY(1) '18'>
        <U2 1 REPLACE(1) '0'>
      >
      <U2 1 TRANSFERSTATE(1) '1'>
      <L [6] TransferInfo(1)
        <A 8 CARRIERID(1) '1FS5012B'>
        <A 5 SOURCEPORT(1) 'A1STK03D_TO01'>
        <A 13 DESTPORT(1) 'A1STK05D_TI01'>
        <U2 1 CSTSIZE(1) '2'>
        <A 13 FINALLOCATION(1) 'A1ELA01J_BP01'>

```

* VEHICLEID will be empty if carrier is on buffer.
CARRIORLOC will be empty if carrier is on vehicle.

```

    <A4 CARRIERTYPE(n) 'TYPE'>
  >
>
...
<L [3]
  <L [3] CommandInfo(n)
    <A 33 COMMANDID(n) '1FS5012B_20100626102925830_MESSYS'>
    <U2 1 PRIORITY(n) '18'>
    <U2 1 REPLACE(n) '0'>
  >
  <U2 1 TRANSFERSTATE(n) '1'>
  <L [6] TransferInfo(1)
    <A 8 CARRIERID(n) '1FS5012B'>
    <A 5 SOURCEPORT(n) 'A1STK06D_TO01'>
    <A 13 DESTPORT(n) 'A1STK08D_TI01'>
    <U2 1 CSTSIZE(n) '2'>
    <A 13 FINALLOCATION(n) 'A1SPT01U_BP01'>
    <A4 CARRIERTYPE(n) 'TYPE'>
  >
>
>
<L [n] UnitAlarmStatList
  <L [6] UnitAlarmInfo(1)
    <A 8 UNITID(1) 'A2LOHS01_0101'>
    <A 4 VEHICLEPORTPOSITION(1) '1001'>
    <A 4 VEHICLENEXTPOSITION(1) '1002'>
    <U4 1 ALARMID(1) '15794431'>
    <A 16 ALARMTEXT(1) 'MAINTENANCE MODE'>
    <U2 1 VEHICLECOMMUNICATIONSTATE(1) '1'>
  >
  ...
  <L [6] UnitAlarmInfo(n)
    <A 8 UNITID(n) 'A2LOHS01_0109'>
    <A 4 VEHICLEPORTPOSITION(n) '1008'>
    <A 4 VEHICLENEXTPOSITION(n) '1009'>
    <U4 1 ALARMID(n) '15794431'>
    <A 16 ALARMTEXT(n) 'MAINTENANCE MODE'>
    <U2 1 VEHICLECOMMUNICATIONSTATE(n) '1'>
  >
>
<L [n] CstInfoList
  <L [2] CstInfo(1)
    <A 8 CARRIERID(1) '1FS6263A'>
    <U2 1 CSTSIZE(1) '2'>
    <A4 CARRIERTYPE(n) 'TYPE'>
  >

```

```

...
<L [2] CstInfo(n)
  <A 8 CARRIERID(n) '1FS6282B'>
  <U2 1 CSTSIZE(n) '2'>
  <A4 CARRIERTYPE(n) 'TYPE'>
>
>
<L [n] EventsEnabled // To report the enabled CEID
  <U2 [1] 1 >
  ...
  <U2 [1] 665 >
>
>

```

2.6. S1F13 (Establish Communications Request)

Stream	Function	Description	Wait Bit	Transfer
1	13	The purpose of this message is to provide a formal means of initializing communications at a logical application level both on power-up and following a break in communications. It should be the following any period where host and Equipment SECS applications are unable to communicate. An attempt to send an Establish Communications. Request (S1,F13) should be repeated at programmable intervals until an Establish Communications Acknowledge(S1,F14) is received within the transaction time-out period with an acknowledgement code accepting the establishment.	1	H ↔ E

1) Definition

None

2) Message Structure

[Case 1 : Equipment → Host]

1. <L [4]

1. <A[6] MDLN> * Model Name of the equipment
2. <A[6] SOFTREV> * Software Revision Number
3. <A[30] EQPNAME> * EQPNAME
4. <U2 MCMD> * Online Mode (OFFLINE, ONLINE)

>

[Case2 : Host → Equipment]

1. <L [0]

>

[Case 1 : Equipment → Host]

MSG:S1F13[Establish Communications Request]

<L [4]

<A[3] MDLN 'RCP'> * 'RCP' is fixed

```

<A[5] SOFTREV '30101'>
<A[30] EQPNAME 'A2LOHS01'>
<U2 MCMD '1'>
>

```

[Case2 : Host → Equipment]

```

MSG:S1F13[Establish Communications Request]
<L [0]
>

```

2.7. S1F14 (Establish Communications Request Acknowledge)

Stream	Function	Description	Wait Bit	Transfer
1	14	Accept or deny Establish Communication Request.	0	H ↔ E

1) Definition

COMMACK

- 0 = ACK
- 1 = Denied, Retry
- 2-63 = Hold

2) Message Structure

[Case 1 : Equipment → Host]

```

1. <L [2]
  1. <B[1] COMMACK>
  2. <L [2]
    1. <A[6] MDLN>          * Model Name of the equipment
    2. <A[6] SOFTREV>       * Software Revision Number
    3. <A[30] EQPNAME>      * EQPNAME
    4. <U2 MCMD>            * Online Mode (OFFLINE, ONLINE)
  >
>

```

[Case2 :Host→ Equipment]

```

1. <L [2]
  1. <B[1] COMMACK>
  2. <L [1]
    1. <A[6] MDLN>          * Model Name of the MCS
  >
>

```

[Case 1 : Equipment → Host]

```

MSG:S1F14[Establish Communications Request Acknowledge]
<L [2]
<B[1] COMMACK '0'>
<L [4]

```

<A[3] MDLN 'RCP'> * 'RCP' is fixed
 <A[5] SOFTREV '30101'>
 <A[30] EQPNAME 'A2LOHS01'>
 <U2 MCMD '1'>

>
 >

[Case2 : Host → Equipment]

<L [2]
 <B[1] COMMACK '0'>
 <L [1]
 <A[3] MDLN 'MCS'>

>
 >

2.8. S1F15 (Request OFF-LINE)

Stream	Function	Description	Wait Bit	Transfer
1	15	The Host requests that the equipment transaction to the OFF-LINE state.	1	H → E

1) Definition

None

2) Message Structure

HEADER ONLY.

MSG:S1F15[Request OFF-LINE]

2.9. S1F16 (OFF-LINE Acknowledge)

Stream	Function	Description	Wait Bit	Transfer
1	16	Acknowledge or Error.	0	H ← E

1) Definition

OFLACK

Acknowledge code for OFF_LINE request

- 0 = Offline Confirmed
- 1 = OFF-LINE Not Allowed
- 2 = Already OFF-LINE
- 3~63 = Reserved

2) Message Structure

1. <B[1] OFLACK>

MSG:S1F16[OFF-LINE Acknowledge]
 <B[1] OFLACK '0'>.

2.10. S1F17 (Request ON-LINE)

Stream	Function	Description	Wait Bit	Transfer
1	17	The Host requests that the equipment transaction to the ON-LINE state.	1	H → E

1) Definition

None

2) Message Structure

HEADER ONLY.

MSG:S1F17[Request ON-LINE]

2.11. S1F18 (ON-LINE Acknowledge)

Stream	Function	Description	Wait Bit	Transfer
1	18	Acknowledge or error	0	H ← E

1) Definition

ONLACK

Acknowledge code for ON_LINE request

0 = ON_LINE Confirmed

1 = Not Allowed to go ON_LINE

2 = Already ON_LINE

3-63 = Hold

2) Message Structure

1. <B[1] ONLACK>

MSG:S1F18[ON-LINE Acknowledge]

<B[1] ONLACK '0'>.

2.12. S2F13 (Equipment Constant Send)

Stream	Function	Description	Wait Bit	Transfer
2	13	Constants such as for calibration, servo gain, alarm limits, data collection mode, and other values that are changed infrequently can be obtained using this message.	1	H → E

1) Definition

* Default mode or this EC data use or not use should be discussed well between equipment maker and SMD

2) Message Structure

1. <L [n]

* n = Number of Equipment Constant

1. <U2 ECID>

n. <U2 ECID>

 $\succ$

* n = 0 means report all ECIDs.

2.13. S2F14 (Equipment Constant Ack)

Stream	Function	Description	Wait Bit	Transfer
2	14	Data Response to S2F13 in the order requested.	0	H ← E

1) Definition

None

2) Message Structure

1. $\leq L[n]$ * n = Number of Equipment Constant

1. $\leq_L [3]$

<A[6] ECV>

<A[16] ECV>

<A[16] ECV>

 $\triangleright$
$$\vdots$$
$$n. < L [3]$$

<A[6] ECV>

<A[16] ECV>

<A[16] ECV>

 $\succ$ $\triangleright$

2.14. S2F15 (New Equipment Constant Send)

Stream	Function	Description	Wait Bit	Transfer
2	15	Change one or more equipment constants.	1	H $\rightarrow$ E

1) Definition

None

2) Message Structure

1. $\leq L[n]$

1. $\leq L [2]$

1. <U2 ECID>

2. $\langle A[n] \text{ ECV} \rangle$ $\succ$

•

$$n. < L [2]$$

1. <U2 ECID>

2. $\langle A[n] \text{ ECV} \rangle$

>
>

2.15. S2F16 (New Equipment Constant Acknowledge)

Stream	Function	Description	Wait Bit	Transfer
2	16	Acknowledge or Error. If EAC contains a non-zero error code, the equipment should not change any of the DVNAME, DVVAL specified in S2F15.	0	H ← E

1) Definition

EAC

Equipment acknowledge code

0 = acknowledge

1 = Denied, At least one constant does not exist

2 = Denied, Busy

3 = Denied, At least one constant out of range

>3 reserved

2) Message Structure

1. <B[1] EAC>

2.16. S2F17 (Date and Time Request)

Stream	Function	Description	Wait Bit	Transfer
2	17	Useful to check equipment time base or for equipment to synchronize with the Host time base	1	H ← E

1) Definition

None

2) Message Structure

HEADER ONLY.

MSG:S2F17[Date and Time Request]

2.17. S2F18 (Date and Time Data)

Stream	Function	Description	Wait Bit	Transfer
2	18	Actual time data.	0	H → E

1) Definition

TIME : [Ascii 16 Byte]

"YYYYMMDDhhmmsscc"

* This time should be used on the equipment controller. This Host time is the standard time for SMD factory's all machine.

2) Message Structure

<A[16] TIME>

MSG:S2F18[Data and Time Request Acknowledge]

<A[16] TIME '2010063013390609' >

2.18. S2F29 (Equipment Constant Name List Request)

Stream	Function	Description	Wait Bit	Transfer
2	29	This function allows the host to retrieve basic information about what equipment constants are available in the equipment.	1	H → E

1) Definition

None

2) Message Structure

1. <L [n] * n = Number of Equipment Constant

1. <U2 ECID>

:

n. <U2 ECID>

>

* n = 0 means report all ECIDs.

2.19. S2F30 (Equipment Constant Name List)

Stream	Function	Description	Wait Bit	Transfer
2	30	Data Response	0	H ← E

1) Definition

None

2) Message Structure

1. <L [n] * n = Number of Equipment Constant

1. <L [6]

1. <U2 ECID>

* Equipment Constant ID

2. <A[64] ECNAME>

* Equipment Constant Name

3. <U2 ECMIN>

* Smallest Equipment Constant

4. <U2 ECMAX>

* Biggest Equipment Constant

5. <U2 ECDEF>

* Specified Equipment Constant

6. <A[32] UNITS>

* Equipment Constant Unit

>

:

n. <L [6]

1. <U2 ECID>

* Equipment Constant ID

2. <A[64] ECNAME>

* Equipment Constant Name

3. <U2 ECMIN>

* Smallest Equipment Constant

4. <U2 ECMAX>

* Biggest Equipment Constant

5. <U2 ECDEF>

* Specified Equipment Constant

6. <A[32] UNITS> * Equipment Constant Unit
>
>

2.20. S2F31 (Date and Time Set Data)

Stream	Function	Description	Wait Bit	Transfer
2	31	Useful to synchronize the equipment time with the host time base.	1	H → E

1) Definition

TIME : [Ascii 16 Byte]

"YYYYMMDDhhmmsscc"

* This time should be used on the equipment controller. This Host time is the standard time for SMD factory's all machine.

2) Message Structure

<A[16] TIME>

MSG:S2F31[Data and Time Set Request]

<A[16] TIME '2010063013390609' >

2.21. S2F32 (Date and Time Set Acknowledge)

Stream	Function	Description	Wait Bit	Transfer
2	32	Acknowledge the receipt of time and date.	0	H ← E

1) Definition

TIACK

Time confirmation code

0 = OK

1 = Error, Not OK

2-63= Hold

2) Message Structure

<B[1] TIACK>

MSG:S2F32[Data and Time Set Request Acknowledge]

<B[1] TIACK '0' >

2.22. S2F33 (Define Report)

Stream	Function	Description	Wait Bit	Transfer
2	33	Host computer specifies a series of event report for equipment.	1	H → E

1) Definition

None

2) Message Structure

```

<L [2]
  1. <U4 DATAID>          * 0 : fixed
  2. <L [n]
    1. <L [2]
      1. <U2 RPTID >      * REPORT ID
      2. <L [m]
        1. <U2 VID>       * Variable ID
        :
        m. <U2 VID>       * Variable ID
      >
    >
    :
  n. <L [2]
    1. <U2 RPTID >      * REPORT ID
    2. <L [m]
      1. <U2 VID>       * Variable ID
      :
      m. <U2 VID>       * Variable ID
    >
  >
>

```

* A list of zero-length following <DATAID> deletes all report definitions and associated links.

* A list of zero-length following <RPTID> deletes report type RPTID. All CEID links to this RPTID are also deleted.

MSG:S2F33[Define Report]

```

<L [3]
  <U4 [1] 0 >          * DATAID
  <L [14]
    <L [2]
      <U2 [1] 1 >      * RPTID = 1
      <L [1]
        <U [1] 61 >    * EqpName
      >
    >
    <L [2]
      <U2 [1] 2 >      * RPTID = 2
      <L [4]
        <U2 [1] 61 >    * EqpName
        <U2 [1] 59 >    * CommandInfo
        <U2 [1] 79 >    * TrasferCompleteInfo
        <U2 [1] 64 >    * ResultCode
      >
    >
    :
  <L [2]

```

```

    <U2 [1] 32 >          * RPTID = 32
    <L [2]
        <U2 [1] 70 >      * VehicleID
        <U2 [1] 58 >      * CommandID
        <U2 [1] 54 >      * CarrierID
        <U2 [1] 377 >     * MovingPointNo
    >
    >
    >
    >

```

2.23. S2F34 (Define Report Acknowledge)

Stream	Function	Description	Wait Bit	Transfer
2	34	Define Report Acknowledge	0	H ← E

1) Definition

DRACK

Define Report Acknowledge code

0 = ACK

1 = Denied. Insufficient space

2 = Denied. Incorrect format

3 = Denied. At least one RPTID is already defined

4 = Denied. At least one VID is already defined

>4 = Some other Error

5-63 = Hold

2) Message Structure

<B[1] DRACK>

MSG:S2F34[Define Report Acknowledge]

<B[1] DRACK '0' >

2.24. S2F35 (Link Event Report)

Stream	Function	Description	Wait Bit	Transfer
2	35	Host computer links a report to CEID	1	H → E

1) Definition

None

2) Message Structure

```

<L [2]
    1. <U4 DATAID >          * 0 : fixed
    2. <L [n]
        1. <L [2]
            1. <U2 CEID >      * Event Id

```


Link Report Acknowledge code

- 0 = ACK
- 1 = Denied. Insufficient space
- 2 = Denied. Incorrect format
- 3 = Denied. At least one CEID link is already defined
- 4 = Denied. There isn't at least one CEID.
- 5 = Denied. There isn't at least one RPTID.
- 6-63 = Reserved

2) Message Structure

<B[1] LRACK>

MSG:S2F36[Link Event Report Acknowledge]

<B[1] LRACK '0' >

2.26. S2F37 (Enable/Disable Event Report)

Stream	Function	Description	Wait Bit	Transfer
2	37	Enable, Disable event report to the host used to send MCS requests only one case CEID	1	H → E

1) Definition

CEED

Collection event or trace enable/disable code

FALSE = Disable

TRUE = Enable

2) Message Structure

<L [2]

1. <BOOL CEED> *Event Enable/Disable Code

2. <L [1]

1. <U2 CEID> *Event ID

>

>

* n = 0 means all CEIDs.

MSG:S2F37[Enable/Disable Event Report]

<L [2]

<BOOLEAN T >

<L [1]

<U2 [1] 3 > * ControlStatusRemote

>

>

2.27. S2F38 (Enable/Disable Event Report Acknowledge)

Stream	Function	Description	Wait Bit	Transfer
--------	----------	-------------	----------	----------

2	38	Acknowledge to Event Enable/Disable Setting	0	H ← E
---	----	---	---	-------

1) Definition

ERACK

Enable/Disable event setting code

0 = ACK

1 = Denied. There isn't at least one CEID.

2-63 = Reserved

2) Message Structure

<B[1] ERACK>

MSG:S2F38[Enable/Disable Event Report Acknowledge]

<B[1] ERACK '0' >

2.28. S2F41 (Host Command Send)

Stream	Function	Description	Wait Bit	Transfer
2	41	The Host Requests the Equipment perform the specified remote command with the associated parameters.	1	H → E

1) Definition

RCMD : [ASCII 20 Bytes]

Remote Command

'ABORT', 'CANCEL', 'PAUSE', 'RESUME', 'UPDATE', 'SCAN', 'INSTALL', 'REMOVE'

2) Message Structure

[ABORT]

<L [2]

1. <A 'RCMD' > * 'ABORT'

2. <L [2]

1. <L [2]

1. <A 'COMMANDID' >

2. <A 'CommandId' >

>

2.<L [2]

1. < A 'REPLACE' >

2. < A 'Replace' > * True or False(Global Duplicate abort or normal abort)

>

>

[REMOVE], [CANCEL]

<L [2]

1. <A 'RCMD' > * REMOVE' or 'CANCEL'

2. <L [1]

1. <L [2]

1. <A 'COMMANDID' >

2. <A 'CommandId' >

>
>
>

[PAUSE], [RESUME]

<L [2]
1. <A 'RCMD' > * 'PAUSE' or 'RESUME'
2. <L [0]
>
>

[UPDATE]

<L [2]
1. <A 'RCMD' > * 'UPDATE'
2. <L [5]
1. <L [2]
1. <A 'COMMANDID' >
2. <A 'CommandID' >
>
2. <L [2]
1. <A 'CARRIERID' >
2. <A 'CarrierID' >
>
3. <L [2]
1. <A 'PRIORITY' >
2. <U2 'Priority' >
>
4. <L [2]
1. <A 'DESTPORT' >
2. <A 'DestPort' >
>
5. <L [2]
1. <A 'FINALLOCATION' >
2. <A 'FianlLocation' >
>
>
>

[SCAN]

<L [2]
<A 'RCMD'> * SCAN
<L [3]
<L [2]
<A 'COMMANDID'>
<A "CommandID">
>
<L [2]
<A 'CARRIERID'>
<A "CarrierID">

```

>
<L [2]
  <A 'CARRIERLOC'>
  <A "CarrierLoc">
>
>
>

```

[INSTALL]

```

<L [2]
  1. <A 'RCMD' >          * 'INSTALL'
  2. <L [4]
    1. <L [2]
      1. <A 'CARRIERID' >
      2. <A "CarrierID" >
    >
    2. <L [2]
      1. <A 'CARRIERLOC' >
      2. <A "CarrierLoc" >
    >
    3. <L [2]
      1. <A 'CSTSIZE' >
      2. <U2 'CstSize' >
    >
    4. <L [2]
      1. <A 'CARRIERTYPE' >
      2. <A "CarrierType" >
    >
  >
>

```

[REMOVE]

```

<L [2]
  1. <A 'RCMD' >          * 'REMOVE'
  2. <L [1]
    1. <L [2]
      1. <A 'CARRIERID' >
      2. <A "CarrierID" >
    >
  >
>

```

[ABORT]

MSG:S2F41[Host Command Send : Abort]

```

<L [2]
  <A 'ABORT' >
  <L [2]

```

```

<L [2]
  <A[9] 'COMMANDID' >
  <A[25] 'A1OHS01D20100630072533499' >
>
<L [2]
  <A[7] 'REPLACE' >
  <A[5] 'FALSE' > * OR TRUE(Normal Abort or Global Duplicate Abort)
>
>
>

```

[CANCEL]

MSG:S2F41[Host Command Send : Cancel]

```

<L [2]
  <A[6] 'CANCEL' >
  <L [1]
    <L [2]
      <A[9] 'COMMANDID' >
      <A[25] 'A1OHS01D20100630072533499' >
    >
  >
>
>

```

[PAUSE], [RESUME]

MSG:S2F41[Host Command Send : Pause]

```

<L [2]
  <A[5] 'PAUSE' >
  <L [0]
>
>

```

MSG:S2F41[Host Command Send : Resume]

```

<L [2]
  <A[6] 'RESUME' >
  <L [0]
>
>

```

[UPDATE]

MSG:S2F41[Host Command Send : Update]

```

<L [2]
  <A[6] 'UPDATE' >
  <L [5]
    <L [2]
      <A[9] 'COMMANDID' >

```

```

    <A[25] 'A1AGVEVD20100630072533499' >
  >
  <L [2]
    <A[9] 'CARRIERID' >
    <A[8] '1FS9063E' >
  >
  <L [2]
    <A[8] 'PRIORITY' >
    <U2[1] 9 >
  >
  <L [2]
    <A[8] 'DESTPORT' >
    <A[13] 'A1EVP02T_IP01' >
  >
  <L [2]
    <A[8] 'FINALLOCATION' >
    <A[13] 'A1COT01D_BP01' >
  >
>
>

```

[SCAN]

MSG:S2F41[Host Command Send : Scan]

```

<L [2]
  <A 'SCAN'>
  <L [3]
    <L [2]
      <A[9] 'COMMANDID'>
      <A[5] "A1234">
    >
    <L [2]
      <A[9] 'CARRIERID'>
      <A[8] '1FS9063E' >
    >
    <L [2]
      <A[10] 'CARRIERLOC'>
      <A[13] 'A1COT01D_BP01' >
    >
  >
>
>

```

[INSTALL]

MSG:S2F41[Host Command Send : Install]

```

<L [2]
  <A[7] 'INSTALL' >
  <L [4]
    <L [2]
      <A[9] 'CARRIERID' >
      <A[8] '1FS9063E' >
    >
    <L [2]
      <A[10] 'CARRIERLOC' >

```

```

    <A[13] 'A1COT01D_BP01' >
  >
  <L [2]
    <A[7] 'CSTSIZE' >
    <U2[1] 0 >
  >
  <L [2]
    <A[13] 'CARRIERTYPE' >
    <A[11] 'CarrierType' >
  >
>
>
>

```

[REMOVE]

MSG:S2F41[Host Command Send : Remove]

```

<L [2]
  <A[6] 'REMOVE' >
  <L [1]
    <L [2]
      <A[9] 'CARRIERID' >
      <A[8] '1FS9063E' >
    >
  >
>
>

```

2.29. S2F42 (Host Command Acknowledge)

Stream	Function	Description	Wait Bit	Transfer
2	42	Acknowledge Host command or error. If command is not accepted due to one or more invalid parameters.	0	H ← E

1) Definition

HCACK

Host Command Parameter Acknowledge Code

- 0 = Confirmed, the command was executed
(The transport system doesn't use this value.
Confirmation is made using the number 4 value.)
- 1 = Command doesn't exist
- 2 = Currently not able to execute
- 3 = At least one parameter isn't valid
- 4 = Confirmed, the command will be executed
and completion will be notified by an event
- 5 = Rejected, already requested
- 6 = Object doesn't exist
- 7 = Already Auto (Only HostCommand(RESUME))

CPACK

Command Parameter Acknowledge Code

0 = No error

1 = CPNAME does not exist

2 = The incorrect value is specified in CPVAL

3 = The incorrect Format is specified in CPVAL

>3 = Another equipment error

4-63 = hold

* These HCACK, CPACK can be expanded the ack type by the discussion afterward.

2) Message Structure

[HCACK != 3] * in the case of no error

<L [2]

1. <B[1] HCACK> * Host Acknowledge

2. <L [0]

>

>

[HCACK = 3] * in the case of a non-valid parameter

<L,2

1. <B[1] HCACK> * Host Acknowledge

2. <L [n] * n = number of parameter groups

1. <L [2]

1. <A[32] CPNAME> * Command Parameter Name

2. <B[1] CPACK > * Control Parameter Ack Code

>

:

n. <L [2]

1. <A[32] CPNAME> * Command Parameter Name

2. <B[1] CPACK > * Control Parameter Ack Code

>

>

>

[HCACK != 3]

MSG:S2F42[Host Command Send Acknowledge]

<L [2]

<B[1] HCACK '4' >

<L [0]

>

>

[HCACK = 3]

MSG:S2F42[Host Command Send Acknowledge]

<L [2]

<B[1] HCACK '3' >

```

<L [2]
  <L [2]
    <A[8] 'PRIORITY' >
    <B[1] 1 >
  >
  <L [2]
    <A[8] 'DESTPORT' >
    <B[1] 1 >
  >
>
>

```

2.30. S2F49 (Enhanced Remote Command)

Stream	Function	Description	Wait Bit	Transfer
2	49	The host requests an object to preform the specified remote command with its associated parameters.	1	H → E

1) Definition

RCMD : [ASCII 20 Bytes]

Remote Command

'TRANSFER',: Send a transfer command,

2) Message Structure

[TRANSFER]

```

<L [4]
  1. <U4 DATAID >          * 0 : fixed
  2. <A[40] OBJSPEC >        * OBJSPEC 'Not Used'
  3. <A[32] RCMD >           * 'TRANSFER'
  4. <L [2]
    1. <L [2]
      1. <A[32] CPNAME >      * 'COMMANDINFO'
      2. <L [3]
        1. <L [2]
          1. <A 'COMMANDID' >
          2. <A "CommandID" >
        >
        2. <L [2]
          1. <A 'PRIORITY' >
          2. <U2 "Priority" >
        >
        3. <L [2]
          1. <A 'REPLACE' >
          2. <U2 "Replace" >
        >
      >
    >
  >

```


Host Command Parameter Acknowledge Code

- . 0 = Confirmed, the command was executed
(The transport system doesn't use this value.
Confirmation is made using the number 4 value.)
- . 1 = Command doesn't exist
- . 2 = Currently not able to execute
- . 3 = At least one parameter isn't valid
- . 4 = Confirmed, the command will be executed
and completion will be notified by an event
- . 5 = Rejected, already requested
- . 6 = Object doesn't exist
- . 7 = Already Auto (Only HostCommand(RESUME))

CPACK

Command Parameter Acknowledge Code

- 0 = No error
- 1 = CPNAME does not exist
- 2 = The incorrect value is specified in CPVAL
- 3 = The incorrect Format is specified in CPVAL
- >3 = Another equipment error
- 4-63 = hold

* These HCACK, CPACK can be expanded the ack type by the discussion afterward.

2) Message Structure

[HCACK != 3]

* in the case of no error

<L [2]

1. <B[1] HCACK> * Host Acknowledge
2. <L [0]

>

>

[HCACK = 3]

* in the case of a non-valid parameter

<L [2]

1. <B[1] HCACK> * Host Acknowledge
2. <L [n] * n = number of parameter groups

1. <L [2]

1. <A [32] CPNAME> * Command Parameter Name
2. <B[1] CPACK > * Control Parameter Ack Code

>

:

n. <L, 2

1. <A [32] CPNAME> * Command Parameter Name
2. <B[1] CPACK > * Control Parameter Ack Code

>

>

>

[HBACK != 3]

MSG:S2F50[Enhanced Remote Command Acknowledge]

```
<L [2]
  <B[1] HBACK '4' >
  <L [0]
>
>
```

[HBACK = 3]

MSG:S2F42[Host Command Send Acknowledge]

```
<L [2]
  <B[1] HBACK '3' >
  <L [2]
    <L [2]
      <A[8] 'PRIORITY' >
      <B[1] 1 >
    >
    <L [2]
      <A[8] 'DESTPORT' >
      <B[1] 1 >
    >
  >
>
>
```

2.32. S5F1 (Alarm Report Send)

Stream	Function	Description	Wait Bit	Transfer
5	1	This message reports a change in or presence of an alarm condition.	1	H ← E

1) Definition

ALCD : [Binary 1 Byte]

Alarm code

bit 8 = 1 generated alarm status

bit 8 = 0 clear alarm status

bit 7~1 is alarm category code

0 = not used

1 = Personal safety

2 = Equipment safety

3 = Parameter control warning

4 = Parameter control error

5 = Irrecoverable error

6 = Equipment state warning

7 = Attention flags

8 = Data integrity

9 = other categories /* This category is related to the light alarm MCS manages, MCS doesn't report this category to MES

ALID : [Unsigned 4 Byte]

Alarm Identification

ALTX : [Ascii 80 Bytes]

Alarm text limited 80 characters.

Possible characters are limited as following :a~z, A~Z, 0~9, _ , space

2) Message Structure

1. <L [4]

1. <B[1] ALCD> * Alarm Code
 2. <U4 ALID> * Alarm ID
 3. <A[80] ALTX> * Alarm Text
 4. <A[30] UNITID> * Alarm UnitID
- >

MSG:S5F1[Alarm Report Send]

<L [4]

<B[1] ALARMCODE '128'>

<U4[1] ALARMID '1441791'>

<A[16] ALARMTEXT 'Maintenance Mode'>

<A[13] UNITID 'A2LOHS01_0101'>

>

2.33. S5F2 (Alarm Report Acknowledge)

Stream	Function	Description	Wait Bit	Transfer
5	2	Acknowledge or error.	0	H → E

1) Definition

ACKC5

0 = ACK

>0 = Error, could not ACK

1-63 = Hold

2) Message Structure

1. <B[1] ACKC5>

MSG:S5F2[Alarm Report Acknowledge]

<B[1] ACKC5 '0' >

2.34. S5F3 (Enable/Disable Alarm Send)

Stream	Function	Description	Wait Bit	Transfer
5	3	This message will change the state of the enable bit in the equipment. The enable bit determines if the alarm will be sent to the host.	1	H → E

1) Definition

None

2) Message Structure

1. <L [2]

1. <B[1] ALED> * Alarm Valid/Invalid Code

2. <U4 ALID> * Alarm ID

>

MSG:S5F3[Enable/Disable Alarm Send]

<L [2]

<B[1] ALARMCODE '128'>

<U4[1] ALARMID '1441791'>

>

2.35. S5F4 (Enable/Disable Alarm Acknowledge)

Stream	Function	Description	Wait Bit	Transfer
5	4	Acknowledge to Alarm Valid/Invalid Setting	0	H ← E

1) Definition

ACKC5

0 = ACK

>0 = Error, could not ACK

1-63 = Hold

2) Message Structure

1. <B[1] ACKC5>

MSG:S5F4[Enable/Disable Alarm Acknowledge]

<B[1] ACKC5 '0' >

2.36. S5F5 (Alarm List Request)

Stream	Function	Description	Wait Bit	Transfer
5	5	Alarm List Request	1	H → E

1) Definition

None

2) Message Structure

1. <L [n]

1. <U4 ALID> * Requested Alarm ID

:

n. <U4 ALID> * Requested Alarm ID

>

* n = 0 means report all alarms regardless of the state of ALED.

MSG:S5F5[Alarm List Request]

<U4 [0] > * 0 : all alarms

2.37. S5F6 (Alarm List Data)

Stream	Function	Description	Wait Bit	Transfer
5	6	Acknowledge to Alarm List Request	0	H ← E

1) Definition

None

2) Message Structure

1. <L [n]

1. <L [3]

1. <B1 ALCD> * Alarm Code
2. <U4 ALID> * Alarm Id
3. <A[80] ALTX> * Alarm Text

>

:

n. <L [3]

1. <B1 ALCD> * Alarm Code
2. <U4 ALID> * Alarm Id
3. <A[80] ALTX> * Alarm Text

>

>

MSG:S5F6[Alarm List Data]

<L [6]

<L [3]

<B[1] ALARMCODE '80'>

<U4[1] ALARMID '941'>

<A[20] 'PORT INTERLOCK ERROR'>

>

<L [3]

<B[1] ALARMCODE '80'>

<U4[1] ALARMID '942'>

<A[16] 'PORT CHUCK ERROR'>

>

<L [3]

<B[1] ALARMCODE '80'>

<U4[1] ALARMID '943'>

<A[19] 'PORT ABNORMAL ERROR'>

>

<L [3]

<B[1] ALARMCODE '80'>

<U4[1] ALARMID '600'>

```

    <A[14] 'Emergency stop' >
  >
  <L [3]
  <B[1] ALARMCODE '80'>
  <U4[1] ALARMID '601'>
    <A[16] 'Safety door open' >
  >
  <L [3]
  <B[1] ALARMCODE '80'>
  <U4[1] ALARMID '610'>
    <A[28] 'Disconnected with master plc' >
  >
>

```

2.38. S6F11 (Event Report Send)

Stream	Function	Description	Wait Bit	Transfer
6	11	The equipment send a defined, linked ,and enabled group of reports to Host upon the occurrence of an event	1	H ← E

1) Definition

None

2) Message Structure

[RPTID = 1]

CEID = 1 :EquipmentOffline
 CEID = 2 :ControlStatusLocal
 CEID = 3 :ControlStatusRemote
 CEID = 10 :MessageRecognition
 CEID = 11 :EstablishIntervalTimeChange
 CEID = 51 :AlarmCleared
 CEID = 52 :AlarmSet
 CEID = 53 :TSCAutoCompleted
 CEID = 54 :TSCAutoInitiated
 CEID = 55 :TSCPauseCompleted
 CEID = 56 :TSCPauseed
 CEID = 57 :TSCPauseInitiated

<L [3]

1. <U4 DATAID>

2. <U2 CEID>

3. <L [1]

1. <L [2]

1. <U2 RPTID> * RPTID = 1

2. <L [1]

1. <AEqpName>

```

>
>
>
>

```

ex1) MSG:S6F11[ControlStateRemote(OnlineRemote)]

```

<L [3]
  <U4 [1] 0 >
  <U2 [1] 3 >          * CEID = 3 → (1, 2, 10, 11, 51, 52, 53, 54, 55, 56, 57)
  <L [1]
    <L [2]
      <U2 [1] 1 >      * RPTID = 1
      <L [1]
        <A [8] "A1OHS01D" >
      >
    >
  >
>

```

[RPTID = 2]

CEID = 107 :TransferCompleted

CEID = 109 :TransferPaused

```

<L [3]
  1. <U4 DATAID>
  2. <U2 CEID>
  3. <L [1]
    1. <L [2]
      1. <U2 RPTID>          * RPTID = 2
      2. <L [5]
        1. <AEqpName>
        2. <L [3]          /* CommandInfo */
          1. <A CommandID>
          2. <U2 Priority>
          3. <U2 Replace>
        >
        3. <L [1]          /* TransferCompleteInf */
          <L [2]
            1. <L [6]
              1. <A CarrierID>
              2. <A SourcePort>
              3. <A DestPort>
              4. <U2 CstSize>
              5. <A FinalLocation>
              6. <A CarrierType>
            >
          >
        >
      >
    >
  >

```

∇

e

/

[1]

CEID = 101 :TransferAbortCompleted
 CEID = 102 :TransferAbortFailed
 CEID = 103 :TransferAbortInitiated
 CEID = 104 :TransferCancelCompleted
 CEID = 105 :TransferCancelFailed
 CEID = 106 :TransferCancelInitiated
 CEID = 108 :TransferInitiated
 CEID = 110 :TransferResumed
 CEID = 111 : Transferring
 CEID = 660 :TransferUpdateCompleted
 CEID = 661 :TransferUpdateFailed
 CEID = 673 :DestUpdateCompleted
 CEID = 674 :PriorityUpdateCompleted

```

<L [3]
  1. <U4 DATAID>
  2. <U2 CEID>
  3. <L [1]
    1. <L [2]
      1. <U2 RPTID>          * RPTID = 3
      2. <L [4]
        1. <AEqpName>
        2. <L [3]          /* CommandInfo */
          1. <A CommandID>
          2. <U2 Priority>
          3. <U2 Replace>
          >
        3. <L [1]          /* TransferCompleteInf */
          <L [2]
            1. <L [6]
              1. <A CarrierID>
              2. <A SourcePort>
              3. <A DestPort>
              4. <U2 CstSize>
              5. <A FinalLocation>
              6. <A CarrierType>
              >
            2. <A CarrirLoc>
            >
          >
        4. <U2 CommandState>
      >
    >
  >
>
  
```

ex1) MSG:S6F11[TransferAbortCompleted]

```
<L [3]
  <U4 [1] 0 >
  <U2 [1] 101 >          * CEID = 101 → (102, 103, 104, 105, 106, 108, 110, 111, 660,
  <L [1]                  661, 673, 674)
    <L [2]
      <U2 [1] 3 >        * RPTID = 3
      <L [4]
        <A [8] "A1OHS01D" >
        <L [3]           /* CommandInfo */
          <A [25] "A1OHS01D20080526072533499" >
          <U2 [1] 5 >
          <U2 [1] 0 >
        >
      <L [1]             /* TransferCompleteInf */
        <L [2]
          <L [6]
            <A [8] "1HS5066C" >
            <A [13] "A1STK02D_TO01" >
            <A [13] "A1STK05D_TI01" >
            <U2 [1] 1 >
            <A [13] "A1ELA04K_BP01" >
            <A [13] "MAGAZINE_S" >
          >
          <A [13] "A1STK02D_TO01" >
        >
      >
    <U2 [1] 3>
  >
>
```

[RPTID = 4]

CEID = 151 :CarrierInstalled

CEID = 152 :CarrierRemoved

```
<L [3]
  1. <U4 DATAID>
  2. <U2 CEID>
  3. <L [1]
    1. <L [2]
      1. <U2 RPTID>      * RPTID = 4
      2. <L [6]
        1. <AEqpName>
        2. <A CommandID>
        3. <A CarrierID>
```

$\succ$

ex1) MSG:S6F11[CarrierInstalled(TSC)]

>

[RPTID = 5]

CEID = 201 :VehicleArrived

CEID = 205 :VehicleDeparted

>
>

ex1) MSG:S6F11[VehicleArrived(TSC)]

```
<L [3]
  <U4 [1] 0 >
  <U2 [1] 201 >          * CEID = 201 → (205)
  <L [1]
    <L [2]
      <U2 [1] 5 >        * RPTID = 5
      <L [4]
        <A [8] "A1OHS01D" >
        <A [25] "A1OHS01D20080526072533499" >
        <A [6] "OHS104" >
        <L [1]
          <A [13] "A1STK03D_TO02" > /* TransferPortList */
        >
      >
    >
  >
>
```

[RPTID = 6]

CEID = 202 :VehicleAcquireStarted
CEID = 203 :VehicleAcquireCompleted
CEID = 206 :VehicleDepositStarted
CEID = 207 :VehicleDepositCompleted
CEID = 255 :CarrierPreHandoff

```
<L [3]
  1. <U4 DATAID>
  2. <U2 CEID>
  3. <L [1]
    1. <L [2]
      1. <U2 RPTID>      * RPTID = 6
      2. <L [5]
        1. <AEqpName>
        2. <A CommandID>
        3. <A VehicleID>
        4. <A CarrierID>
        5. <A TransferPort>
      >
    >
  >
>
```

ex1) MSG:S6F11[VehicleAcquireStarted(TSC)]

```
<L [3]
  <U4 [1] 0 >
  <U2 [1] 202 >          * CEID = 202 → (203, 206, 207, 255)
  <L [1]
    <L [2]
      <U2 [1] 6 >          * RPTID = 6
      <L [5]
        <A [8] "A1OHS01D" >
        <A [25] "A1OHS01D20080526072533499" >
        <A [6] "OHS104" >
        <A [8] "1HS5074B" >
        <A [13] "A1STK04D_TO01" >
      >
    >
  >
>
```

[RPTID = 7]

CEID = 204 :VehicleAssigned

CEID = 210 :VehicleUnassigned

```
<L [3]
  1. <U4 DATAID>
  2. <U2 CEID>
  3. <L [1]
    1. <L [2]
      1. <U2 RPTID>          * RPTID = 7
      2. <L [3]
        1. <AEqpName>
        2. <A VehicleID>
        3. <A CommandID>
      >
    >
  >
>
```

ex1) MSG:S6F11[VehicleAssigned]

```
<L [3]
  <U4 [1] 0 >
  <U2 [1] 204 >          * CEID = 204 → (210)
  <L [1]
    <L [2]
      <U2 [1] 7 >          * RPTID = 7
      <L [3]
        <A [8] "A1OHS01D" >
```

```

    <A [6] "OHS104" >
    <A [25] "A1OHS01D20080829145766014" >
  >
>
>
>
>

```

[RPTID = 8]

CEID = 208 :VehicleInstalled

CEID = 209 :VehicleRemoved

```

<L [3]
  1. <U4 DATAID>
  2. <U2 CEID>
  3. <L [1]
    1. <L [2]
      1. <U2 RPTID>          * RPTID = 8
      2. <L [2]
        1. <AEqpName>
        2. <A VehicleID>
      >
    >
  >
>

```

ex1) MSG:S6F11[VehicleInstalled]

```

<L [3]
  <U4 [1] 0 >
  <U2 [1] 208 >          * CEID = 208 → (209)
  <L [1]
    <L [2]
      <U2 [1] 8 >          * RPTID = 8
      <L [1]
        <A [8] "A1OHS01D" >
        <A [6] "OHS101" >
      >
    >
  >
>

```

[RPTID = 9]

CEID = 254 :OperatorInitiatedAction

```

<L [3]
  1. <U4 DATAID>
  2. <U2 CEID>
  3. <L [1]

```

```

1. <L [2]
  1. <U2 RPTID> * RPTID = 9
  2. <L [7]
    1. <A CommandID>
    2. <A CommandType>
    3. <A CarrierID>
    4. <A SourcePort>
    5. <A DestPort>
    6. <U2 Priority>
    6. <U2 CarrierCleanStatus>
      >
    >
  >
>

```

ex1) MSG:S6F11[OperatorInitiatedAction]

```

<L [3]
  <U4 [1] 0 >
  <U2 [1] 254 > * CEID = 254
  <L [1]
    <L [2]
      <U2 [1] 9 > * RPTID = 9
      <L [6]
        <A [13] "MNL0905143922" >
        <A [13] "EQ_TRANSFER" >
        <A [8] "1HS5074B" >
        <A [13] "A1STK03D_TO01" >
        <A [13] "A1STK04D_TI01" >
        <U2 [1] 3 >
        <U2 N>
      >
    >
  >
>

```

[RPTID = 10]

CEID = 260 :PortOutOfService

CEID = 261 :PortInService

```

<L [3]
  1. <U4 DATAID>
  2. <U2 CEID>
  3. <L [1]
    1. <L [2]
      1. <U2 RPTID> * RPTID = 10

```

```

2. <L [2]
  1. <AEqpName>
  2. <A PortID>
>
>
>
>
>

```

ex1) MSG:S6F11[PortOutOfService]

```

<L [3]
  <U4 [1] 0 >
  <U2 [1] 260 >          * CEID = 260 → (261)
  <L [1]
    <L [2]
      <U2 [1] 10 >      * RPTID = 10
      <L [1]
        <A [8] "A1OHS01D" >
        <A [6] "A1STK01D_TO01" >
      >
    >
  >
>
>

```

[RPTID = 11]

CEID = 251 :CarrierIDRead

```

<L [3]
  1. <U4 DATAID>
  2. <U2 CEID>
  3. <L [1]
    1. <L [2]
      1. <U2 RPTID>          * RPTID = 11
      2. <L [5]
        1. <AEqpName>
        2. <A CommandID>
        3. <A CarrierID>
        4. <U2 IDReadStatus> * 0 : Success, 1 : Fail, 2 : Duplicate, 3 : Mismatch
        5. <A CarrierLoc>
      >
    >
  >
>

```

ex1) MSG:S6F11[CarrierIDRead]

```

<L [3]

```

```

<U4 [1] 0 >
<U2 [1] 251 >          * CEID = 251
<L [1]
  <L [2]
    <U2 [1] 11 >        * RPTID = 11
    <L [5]
      <A [8] "A1OHS01D" >
      <A [25] "A1OHS01D20080526072533499" >
      <A [8] "1HS5066C" >
      <U2 [1] 1 >        * 0 : Success, 1 : Fail, 2 : Duplicate, 3 : Mismatch
      <A [10] "CARRIERLOC">
    >
  >
>
>

```

[RPTID = 13]

CEID = 503 :UnitAlarmCleared

CEID = 504 :UnitAlarmSet

```

<L [3]
  1. <U4 DATAID>
  2. <U2 CEID>
  3. <L [1]
    1. <L [2]
      1. <U2 RPTID>      * RPTID = 13
      2. <L [6]
        1. <AEqpName>
        2. <A UnitID>
        3. <U4 AlarmID>
        4. <AAlarmText>
        5. <A VehiclePortPosition>
        6. <A UnitStatusCleable>
      >
    >
  >
>

```

ex1) MSG:S6F11[UnitAlarmCleared]

```

<L [3]
  <U4 [1] 0 >
  <U2 [1] 503 >          * CEID = 503 → (504)
  <L [1]
    <L [2]
      <U2 [1] 13 >        * RPTID = 13
      <L [6]

```

```

    <A [8] "A1OHS01D" >
    <A [6] "OHS103" >
    <U4 [1] 8781823 >
    <A [16] "Maintenance Mode" >
    <A [1] "6009" >
    <A [1] "Y" >
  >
>
>
>
>

```

[RPTID = 14]

CEID = 270 :MonitoredVehicles

```

<L [3]
  1. <U4 DATAID>
  2. <U2 CEID>
  3. <L [1]
    1. <L [2]
      1. <U2 RPTID> * RPTID = 14
      2. <L [7]
        1. <A VehicleID>
        2. <A VehicleCurrentPosition>
        3. <A VehicleNextPosition>
        4. <U4 AlarmID>
        5. <AAAlarmText>
        6. <U2 VehicleCommunicationState>
        7. <U2 MainteState>
      >
    >
  >
>

```

ex1) MSG:S6F11[MonitoredVehicles]

```

<L [3]
  <U4 [1] 0 >
  <U2 [1] 270 > * CEID = 270
  <L [1]
    <L [2]
      <U2 [1] 14 > * RPTID = 14
      <L [7]
        <A [6] "OHS103" >
        <A [4] "2102" >
        <A [4] "2103" >
        <U4 [1] "6009" >
        <A [16] "Maintenance Mode" >
      >
    >
  >

```

```

    <U2 [1] "1" >
    <U2 [1] "0" >
  >
>
>
>
>
>
>
>

```

[RPTID = 16]

CEID = 153 :VehicleCarrierInstalled

CEID = 154 :VehicleCarrierRemoved

```

<L [3]
  1. <U4 DATAID>
  2. <U2 CEID>
  3. <L [1]
    1. <L [2]
      1. <U2 RPTID>          * RPTID = 16
      2. <L [8]
        1. <AEqpName>
        2. <A CommandID>
        3. <A CarrierID>
        4. <A CarrierLoc>
        5. <A VehicleID>
        6. <U2 CstSize>
        7. <A CarrierType>
        8. <U2 CstPosition>
      >
    >
  >
>

```

ex1) MSG:S6F11[VehicleCarrierInstalled(TSC)]

```

<L [3]
  <U4 [1] 0 >
  <U2 [1] 153 >          * CEID = 153
  <L [1]
    <L [2]
      <U2 [1] 4 >          * RPTID = 16
      <L [8]
        <A [8] "A1OHS01D" >
        <A [25] "A1OHS01D20080526072533499" >

```

```

    <A [8] "1HS5074B" >
    <A [6] "OHS104" >
    <A [6] "OHS104" >
    <U2 [1] 1 >
    <A [11] "CARRIERTYPE">
    <U2 [1] 1>
  >
>
>
>
>

```

[RPTID = 32]

CEID = 671 :VehicleBlockigStarted

CEID = 672 :VehicleBlockigEnded

```

<L [3]
  1. <U4 DATAID>
  2. <U2 CEID>
  3. <L [1]
    1. <L [2]
      1. <U2 RPTID>          * RPTID = 32
      2. <L [4]
        1. <A VehicleID>
        2. <A CommandID>
        3. <A CarrierID>
        4. <A MovingPointNo>
      >
    >
  >
>
>
>

```

ex1) MSG:S6F11[VehicleMovingStarted]

```

<L [3]
  <U4 [1] 0 >
  <U2 [1] 671 >          * CEID = 671 → (672)
  <L [1]
    <L [2]
      <U2 [1] 32 >      * RPTID = 32
    <L [4]
      <A [8] "AGV101" >
      <A [25] "A1OHS01D20080526072533499" >
      <A [8] "1HS5066C" >
      <A [6] "123" >
    >
  >
>

```

>
>

[AGV ONLY]

[RPTID = 31]

CEID = 662 :VehicleChargeSTStarted

CEID = 663 :VehicleChargeFailed

CEID = 664 :VehicleCharging

CEID = 665 :VehicleNotCharged

<L [3]

1. <U4 DATAID>

2. <U2 CEID>

3. <L [1]

1. <L [2]

1. <U2 RPTID> * RPTID = 31

2. <L [2]

1. <A VehicleID>

2. <A ChargePortNo>

>

>

>

>

ex1) MSG:S6F11[VehicleChargeSTStarted]

<L [3]

<U4 [1] 0 >

<U2 [1] 662 > * CEID = 662 → (663, 664, 665)

<L [1]

<L [2]

<U2 [1] 31 > * RPTID = 31

<L [1]

<A [8] "AGV101" >

<A [6] "123" >

>

>

>

>

[Manual Port ONLY]

[RPTID = 33]

CEID = 158 : CarrierWaitIn

<L [3]

1. <U4 DATAID>

2. <U2 CEID>

```

3. <L [1]
  1. <L [2]
    1. <U2 RPTID> * RPTID = 33
    2. <L [5]
      1. <A EqpName>
      2. <A CommandID>
      3. <A CarrierID>
      4. <A CarrierLoc>
      5. <U2 CstSize>
    >
  >
>

```

ex1) MSG:S6F11[CarrierWaitIn(TSC)]

```

<L [3]
  <U4 [1] 0 >
  <U2 [1] 202 > * CEID = 158
  <L [1]
    <L [2]
      <U2 [1] 6 > * RPTID = 33
      <L [5]
        <A [8] "A2EMCT01" >
        <A [25] "2OM0108G20080526072533499_MESSYS" >
        <A [8] "1HS5074B" >
        <A [13] "A2EMCT01_TO01" >
        <U2 [1] "1" >
      >
    >
  >
>

```

[RPTID = 34]

CEID = 159 : CarrierWaitOut

```

<L [3]
  1. <U4 DATAID>
  2. <U2 CEID>
  3. <L [1]
    1. <L [2]
      1. <U2 RPTID> * RPTID = 34
      2. <L [5]
        1. <A EqpName>
        2. <A CommandID>
        3. <A CarrierID>
        4. <A CarrierLoc>

```

```

5. <A PortType>
>
>
>
>

```

ex1) MSG:S6F11[CarrierWaitOut(TSC)]

```

<L [3]
  <U4 [1] 0 >
  <U2 [1] 202 >          * CEID = 159
  <L [1]
    <L [2]
      <U2 [1] 6 >        * RPTID = 34
      <L [5]
        <A [8] "A2EMCT01" >
        <A [25] "2OM0108G20080526072533499_MESSYS" >
        <A [8] "1HS5074B" >
        <A [13] "A2EMCT01_TO01" >
        <A [2] "LP" >
      >
    >
  >
>

```

2.39. S6F12 (Event Report Acknowledge)

Stream	Function	Description	Wait Bit	Transfer
6	12	Acknowledge or error.	0	H → E

1) Definition

ACKC6

0 = ACK

>0 = Error, could not ACK

1-63 = Hold

2) Message Structure

1. <B[1] ACKC6>

MSG:S6F12[Event Report Acknowledge]

<B[1] ACKC6 '0' >

2.40. S6F15 (Event Report Request)

Stream	Function	Description	Wait Bit	Transfer
--------	----------	-------------	----------	----------

6	15	The purpose of this message is for the host to demand a given report group from the equipment.	1	H → E
---	----	--	---	-------

1) Definition

None

2) Message Structure

1. <U2 CEID>

MSG:S6F15[Event Report Request]

<U2[1] 3 > * ControlStatusRemote

2.41. S6F16 (Event Report Data)

Stream	Function	Description	Wait Bit	Transfer
6	16	Equipment sends reports linked to given CEID to host.	0	H ← E

1) Definition

None

2) Message Structure

Same as Event Report Send

MSG:S6F16[Event Report Data]  ACK Data of S6F15

<L [3]

<U4 [1] 0 >

<U2 [1] 3 > * CEID = 3

<L [1]

<L [2]

<U2 [1] 1 > * RPTID = 1

<L [1]

<A [8] "A1OHS01D" >

>

>

>

>

2.42. S9F1 (Unrecognized Device ID)

Stream	Function	Description	Wait Bit	Transfer
9	1	The device ID in the message block header did not correspond to any known device ID in the node detection the error.	0	H ← E

1) Definition

MHEAD : [Binary 10 Bytes]

SECS message block header associated with message block in error

2) Message Structure

<MHEAD>

2.43. S9F3 (Unrecognized Stream Type)

Stream	Function	Description	Wait Bit	Transfer
9	3	The equipment does not recognize the stream type in the message block header.	0	H ← E

1) Definition

MHEAD : [Binary 10 Bytes]

SECS message block header associated with message block in error

2) Message Structure

<MHEAD>

2.44. S9F5 (Unrecognized Function Type)

Stream	Function	Description	Wait Bit	Transfer
9	5	This message indicates that the function in the message ID is not recognized by the receiver.	0	H ← E

1) Definition

MHEAD : [Binary 10 Bytes]

SECS message block header associated with message block in error

2) Message Structure

<MHEAD>

2.45. S9F7 (Illegal Data)

Stream	Function	Description	Wait Bit	Transfer
9	7	This message indicates that the stream and function were recognized but the associated data format could not be interpreted.	0	H ← E

1) Definition

MHEAD : [Binary 10 Bytes]

SECS message block header associated with message block in error

2) Message Structure

<MHEAD>

2.46. S9F9 (Transaction Timer time-out)

Stream	Function	Description	Wait Bit	Transfer
9	9	This message indicates that a transaction(receive) timer has timed out and that the corresponding transaction has been aborted.	0	H ← E

1) Definition

SHEAD : [Binary 10 Bytes]

Stored header related to the transaction timer

2) Message Structure

<SHEAD>